
3AL Hand-in 2 2024

Due: Monday 8 April 11:00 pm

Please make sure all of your answers are clearly labelled.

1. Determine whether each statement is true or false. No explanation needed.
 - (a) Let X be a G -set and $x \in X$. Then $x \in X_f$ if and only if $|G : S(x)| = 1$.
 - (b) If G is a group acting on a set X , then $|G| > |X|$.
 - (c) If X is a G -set with $x \in X$, then $G * x$ is a subgroup of $S(x)$.
 - (d) Let G be a group. If the group action of conjugation (by G on G) is the trivial group action, then G is abelian.

[4 marks]

2. Let G be a group and $\text{Aut}(G)$ be the group of automorphisms of G .
 - (a) Prove that $*$: $\text{Aut}(G) \times G \rightarrow G$ defined by $\varphi * g = \varphi(g)$ is a group action.
 - (b) Find $S(1_G)$.

[3 + 1 = 4 marks]

3. Let G be a finite group acting on a set X and let $x \in X$. Prove or disprove: The number of elements in $G * x$ divides $|G|$.

[4 marks]

Maximum mark = 10

Available marks = 12